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Pages: 2

Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)



Course Code: MAT201

Course Name: Partial Differential equations and Complex analysis

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

- 1 Derive a partial differential equation from the relation $z = (x + y) f(x^2 - y^2)$ (3)
- 2 Solve using direct integration $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$ (3)
- 3 Solve $2z = xp + yq$. (3)
- 4 Write any three assumptions in deriving one dimensional heat equation. (3)
- 5 Show that an analytic function $f(z) = u + iv$ is constant if its real part is constant. (3)
- 6 Show that the function $u = \sin x \cos hy$ is harmonic. (3)
- 7 Find the Maclaurin series of $f(z) = \sin z$ (3)
- 8 Evaluate $\oint_C \ln z \, dz$, where C is the unit circle $|z| = 1$. (3)
- 9 Find all singular points and residue of the function $\operatorname{cosec} z$ (3)
- 10 Determine the location and order of zeros of the function $\sin^4\left(\frac{z}{2}\right)$ (3)

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11 (a) Form the Partial differential equation by eliminating the arbitrary constants from $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$ (5)
- (b) Solve $2xz - p x^2 - 2qxy + pq = 0$ (9)
- 12 (a) Solve $\frac{\partial^3 z}{\partial^2 x \partial y} = \cos(2x + 3y)$ (7)
- (b) Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$ (7)

Module 2

- 13 (a) Derive the solution of the one dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ using variable separable method. (6)
- (b) An insulated rod of length l has its ends A and B maintained at 0°C and

100° C respectively until steady state conditions prevail. If B is suddenly reduced to 0° C and maintained at 0° C, find the temperature at a distance x from A at time t . (8)

14 (a) Derive the one dimensional heat flow equation. (6)

(b) A tightly stretched string of length l with fixed ends is initially in equilibrium position. If it is set vibrating by giving each point a velocity $v_0 \sin^3\left(\frac{\pi x}{l}\right)$. Find the displacement $y(x, t)$. (8)

Module 3

15 (a) Find an analytic function whose real part is $u = \sin x \cosh y$ (7)

(b) Find the image of the strip $\frac{1}{2} \leq x \leq 1$ under the transformation $w = z^2$ (7)

16 (a) Check whether $w = \log z$ is analytic. (8)

(b) Show that under the transformation $w = \frac{1}{z}$, the circle $x^2 + y^2 - 6x = 0$ is transformed into a straight line in the W plane. (6)

Module 4

17 (a) Integrate counter clockwise around the unit circle $\oint_C \frac{\sin 2z}{z^4} dz$ (7)

(b) Find the Taylor series of $\frac{1}{1+z}$ about the centre $z_0 = i$ (7)

18 (a) Evaluate $\int_0^{1+i} (x - y + ix^2) dz$ along the parabola $y = x^2$. (7)

(b) Evaluate $\oint_C \frac{\log z}{(z-4)^2} dz$ counter clockwise around the circle $|z - 3| = 2$. (7)

Module 5

19 (a) Find the Laurent's series expansion of $\frac{z^2 - 1}{z^2 - 5z + 6}$ about $z = 0$ in the region (5)

$2 < |z| < 3$
(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta}$. (9)

20 (a) Evaluate $\oint_C \frac{z-23}{z^2-4z-5} dz$ where $C : |z - 2 - i| = 3.2$ using Residue theorem. (5)

(b) Evaluate $\int_0^\infty \frac{(x^2 + 2)dx}{(x^2 + 1)(x^2 + 4)}$. (9)
